

Lab 3 Codec / Adversarial Generation Summary

Deep Generative Genre Remastering

Lab 3 converts the frozen analysis stack from Labs 1 and 2 into a short-form conditional generator. The codec branch does not generate audio from pure noise. Instead, it performs adversarial latent translation inside a pretrained EnCodec space. A source waveform is encoded into latent codes q_{src} , Lab 1 supplies z_{content} as the structural anchor, and a target style embedding supplies the genre condition. A FiLM-conditioned translator predicts edited or rewritten codec latents q_{hat} , and a frozen EnCodec decoder reconstructs the waveform. A multi-scale discriminator, feature-matching loss, content loss, and spectral losses are then used to keep the translated output plausible while preserving melody. The branch is therefore best described as **adversarial source-conditioned editing** rather than generation from scratch.

Three architectural changes were especially important. First, EnCodec was used as a decoder guardrail so that the generator would stay on a realistic audio manifold. Second, MERT-probe embeddings replaced weaker style centroids, which improved conditioning and separated target styles more clearly. Third, direct-output translation was enabled so the translator could move beyond a small residual around the source latent code. These changes were responsible for the first large jump in Lab 3 style metrics.

The main metrics in this report serve different purposes. **MPS** (melodic preservation score) measures cosine similarity between source and generated content embeddings, so higher values mean the original melodic and structural identity survived. **Style confidence** is the mean probability that the internal Lab 3 judge assigns to the target genre, while **style accuracy** is the fraction of clips whose target genre is predicted correctly; together they describe how strongly the internal judge believes style transfer has happened. **Target accuracy** and **target cosine** are generated-audio realism-supervisor measures computed on held-out transfers, so they are a stricter test of whether the output actually moved toward the target style rather than merely satisfying the training-time judge. **FAD** is Fréchet Audio Distance over pretrained MERT embeddings, where lower is better and smaller values indicate that generated audio sits closer to real audio. **HF MAE** is the absolute error in high-frequency energy relative to target-reference audio, so smaller values usually mean less brittle, screechy, or over-bright output.

The first clearly successful codec run was **run1055**. It became the original best Lab 3 run because it achieved strong internal gate metrics while preserving melody:

Run	MPS	Style conf.	Style acc.	Pairwise cos.
run1055	0.9565	0.8940	0.9492	0.0612
run1056	0.9541	0.6534	0.7266	0.1634
run1057probe4	0.9520	0.1447	0.1367	0.3955
codec_reset_20260310_051425	0.9571	0.8943	0.9531	0.0426

However, later analysis showed that these internal gate metrics overstated how convincing the generated outputs really were. The realism supervisor measured generated-audio realism and target-style movement directly, and that changed the interpretation of the runs:

Run	MPS	Target acc.	Target cos.	FAD	HF MAE
run1055	0.9991	0.2500	-0.0998	27.01	0.1760
run1056	0.9996	0.0833	-0.1572	17.92	0.1039
run1057probe4	0.9888	0.3750	0.1655	41.84	0.1309
codec_reset_20260310_051425	0.9853	0.6250	0.5956	35.07	0.0590

These numbers show the main pattern observed during Lab 3. **run1055** was the strongest early success under the original Lab 3 gate, but its generated-audio target-style movement was weaker than the gate implied. **run1056** improved realism substantially and reduced harshness, but style movement collapsed, making the outputs safer but less convincingly remastered. **run1057probe4** restored some target-style movement with

stronger late-stage tuning, but realism degraded too far. The best overall balance was the overnight reset run `codec_reset_20260310_051425`, which combined strong Lab 3 gate performance with the best generated-audio style movement seen so far in the codec branch, although its Fréchet distance still missed the stricter realism threshold.

The main conclusion from Lab 3 is that the codec/GAN branch is effective as a stable short-form remastering system. It preserves melody very well, it benefits strongly from adversarial training, and it can produce noticeable style transfer when the style-conditioning and direct-output settings are tuned correctly. At the same time, the branch remains structurally biased toward source-conditioned editing rather than full re-authoring. This is why realism and style movement still trade off against one another. The branch is therefore a successful and important part of the project, but it is better understood as a high-quality adversarial editor than as a complete from-scratch genre generator.

Primary run artifacts used in this summary: `run_state.json` and `codec_realism_best.json` from `run1055`, `run1056`, `run1057probe4`, and `codec_reset_20260310_051425`.